

Baselines

Persistence, EMA mean and the adaptive RLS AR(8) reference.

Baselines need no training. Every candidate is reported against rls_ar and persistence; a model that does not beat rls_ar is not progress.

baseline	definition	cost/beat	val open	10%	30%	50%	80%	test whole-series 80%
persistence	next dRR = 0 (repeat the last RR)	0	41.70	31.28	33.65	37.13	46.42	48.55
ema_mean	next RR = exponential moving average of the window (factor 0.75 per beat), dRR = EMA - last RR	30	43.58	33.89	35.89	38.40	43.79	46.78
rls_ar	adaptive per-patient AR(8) by recursive least squares on centred RR lags (forgetting 0.99); the forecast is the rls_pred feature	152 FLOPs	37.14	28.88	33.11	37.47	53.73	49.15

RMSE in ms. **val open loop**: one-step-ahead on the true history, 16 validation subjects. **10-80%**: closed loop (pacemaker use) on validation: that share of beats dropped at random from beat 0, each filled by the model and fed back; RMSE on model-filled beats, first 5000 beats per subject, mean of the seeds of that round (3 seeds from round 2 on). int8 is used where it exists. **RSA invented**: share of subject-runs whose filled series gains an RSA peak the original lacks. **test whole-series 80%**: diagnostic on the 10 test subjects (whole series, seed 7, results/test/ folder); it was not used for any selection.

Correlation between the original and the filled series (validation and test subjects)

model	corr RR validation	corr RR test	corr dRR validation	corr dRR test	RMSE filled, validation (ms)	RMSE filled, test (ms)
persistence	0.9221	0.9260	0.216	0.213	56.05	48.55
ema_mean	0.9289	0.9328	0.243	0.228	52.67	46.78
rls_ar	0.9186	0.9276	0.202	0.219	56.34	49.15

Computed on the 16 validation and the 10 test subjects (results/validation/ and results/test/ folders), on the ENTIRE series, after dropping 80% of the beats (seed 7, from beat 0) and filling them in closed loop with the model (Burg-AR fill for the first 30 beats): **corr RR** = np.corrcoef(original_serie, modified_serie)[0, 1]; **corr dRR** = the same function on the beat-to-beat changes, np.corrcoef(np.diff(original), np.diff(filled))[0, 1], which isolates the short-term dynamics. Mean over subjects; per-subject values in <folder>/crosscorr.csv. **RMSE filled**: RMSE on the model-filled beats of the same runs. Diagnostic: the selection used the round-4 protocol, not these runs.

Observations

- rls_ar is the strongest baseline one step ahead (37.1 ms on validation) and is the residual base of every good model.
- In the closed loop rls_ar is unstable at high drop rates: at 80% dropped it is worse than persistence and invents an RSA peak in about 75% of subject-runs, because an AR model fed its own predictions resonates. Freezing its update on filled beats made it worse (tested in round 4).
- persistence never invents RSA peaks but damps RSA power and has the worst one-step error.

Abbreviations and acronyms used in this document

term	meaning
AR	autoregressive: a value predicted as a weighted sum of past values; AR(p) uses p past values
ar	feature-set name: autoregressive inputs (rr_dev and the 8 most recent dRR) of the linear model
Burg-AR	AR model whose coefficients were fitted with Burg's method (used for the first 30 beats)
dRR	beat-to-beat change of the RR interval: $dRR[n+1] = RR[n+1] - RR[n]$ (ms); the quantity predicted
EMA	exponential moving average
ema_mean	baseline predicting the next RR as an exponential moving average of the window
FLOPs	floating-point operations (arithmetic cost)
HRV	heart rate variability
int8	8-bit integer: weights and activations quantised to 8-bit integers, as run on the microcontroller; _int8 = the int8 version of a run
ms	milliseconds
np.corrcoef	NumPy's Pearson correlation-coefficient function
np.diff	NumPy's successive-difference function
RLS	recursive least squares: an adaptive filter that updates its coefficients at every beat
rls_ar	baseline: adaptive per-patient AR(8) forecast by RLS
rls_pred	feature: RLS AR(8) forecast of the next dRR (ms), clamped to +/-300
RMSE	root mean square error (ms): the main accuracy measure, lower is better
RR	RR interval: time between two consecutive heartbeats (R waves of the ECG), in ms
RSA	respiratory sinus arrhythmia: the oscillation of the heart rate driven by breathing
val	validation: the 16 validation subjects, used for model selection